

Linux Users, Groups and File Permissions

Detailed Classroom Notes

Introduction

Linux is a multi-user operating system. Multiple users can work on the same machine simultaneously.

To maintain security, Linux uses Users, Groups and Permissions. Together they determine who can access files, modify files, execute programs and administer the system.

Users in Linux

A user is an account that can log in to the system. Every user is identified using a unique User ID (UID).

Types of users include Root User, Regular Users and System Users. Root has complete administrative privileges

while regular users have limited permissions. System users are used by services such as web servers and databases.

Important User Files

User information is stored in `/etc/passwd` while encrypted password information is stored in `/etc/shadow`.

A typical `passwd` entry looks like:

```
sameer:x:1000:1000:Sameer:/home/sameer:/bin/bash
```

Fields represent username, password placeholder, UID, GID, comment field, home directory and login shell.

User Management Commands

`whoami` - Display current user

`who` - Display logged-in users

`w` - Detailed logged-in user information

`id` - Show UID, GID and groups

`useradd john` - Create a user

`passwd john` - Set password

`usermod -aG dev john` - Add user to group

`userdel -r john` - Delete user and home directory

Groups in Linux

A group is a collection of users. Instead of assigning permissions individually to each user, permissions can be assigned to groups. Every user has one Primary Group and may belong to multiple Secondary Groups.

Group Commands

`groupadd developers`

`groupdel developers`

`groups john`

`usermod -aG developers john`

File Ownership

Every file and directory in Linux has an Owner and a Group.

Example:

```
-rw-r--r-- 1 john developers 500 Jun 8 file.txt
```

Owner = john

Group = developers

Others = Everyone else on the system

Commands:

```
chown sameer file.txt
```

```
chown sameer:developers file.txt
```

```
chgrp developers file.txt
```

Understanding Permissions

Linux permissions are divided into three categories:

User (Owner)

Group

Others

Each category can have:

Read (r)

Write (w)

Execute (x)

Example:

```
-rwxr-xr--
```

Owner = rwx

Group = r-x

Others = r--

Permission Meanings

Read (4): View file contents.

Write (2): Modify file contents.

Execute (1): Run a file as a program or script.

For directories:

Read = View filenames.

Write = Create/Delete files.

Execute = Enter the directory using cd.

Numeric Permissions

$rwx = 7 (4+2+1)$

$rw- = 6 (4+2)$

$r-x = 5 (4+1)$

$r-- = 4$

$--- = 0$

Common permissions:

777 = rwxrwxrwx
755 = rwxr-xr-x
700 = rwx-----
644 = rw-r--r--
600 = rw-----

chmod Examples

```
chmod 777 file.txt # Full permissions
chmod 755 script.sh # Typical executable script
chmod 600 secrets.txt # Private file
```

Symbolic Mode:

```
chmod +x script.sh
chmod g+w project.txt
chmod o-r confidential.txt
chmod a+x install.sh
```

Recursive Operations

Apply changes to an entire directory structure:

```
chmod -R 755 project
chown -R john:developers project
```

These commands are frequently used by Linux administrators.

Practical Example

Step 1: Create users

```
sudo useradd john
sudo useradd mike
```

Step 2: Create a group

```
sudo groupadd developers
```

Step 3: Add users to the group

```
sudo usermod -aG developers john
sudo usermod -aG developers mike
```

Step 4: Create a file

```
touch project.txt
```

Step 5: Assign ownership

```
sudo chown john:developers project.txt
```

Step 6: Assign permissions

```
chmod 664 project.txt
```

Result:

Owner = Read + Write

Group = Read + Write

Others = Read Only

Important Commands Summary

whoami, who, w, id, groups,
useradd, userdel, usermod, passwd,
groupadd, groupdel,
ls -l, chmod, chown, chgrp,
sudo

Memory Trick

Think of every Linux file as having three locks:

Owner Lock

Group Lock

Others Lock

Each lock can have three keys:

Read (4)

Write (2)

Execute (1)

Example:

chmod 754 file.txt

Owner = 7 = rwx

Group = 5 = r-x

Others = 4 = r--

This concept forms the foundation of Linux administration, DevOps, Cloud Computing and Cybersecurity.

Practice Questions

1. What is the difference between a root user and a regular user?
2. What is the purpose of groups in Linux?
3. Explain the meaning of 755 and 644 permissions.
4. What is the difference between chmod, chown and chgrp?
5. Explain directory permissions with examples.
6. Write commands to create a user and assign a password.